

MAT 426, Class 19: Quadratic Forms & Singular Value Decomposition

PRINCIPAL AXES THEOREM Let A be an $n \times n$ _____ matrix. Then, there exists an _____ change of variable, $\mathbf{x} = P\mathbf{y}$ that transforms the quadratic form $\mathbf{x}^T A \mathbf{x}$ into a quadratic form $\mathbf{y}^T D \mathbf{y}$ with _____ term.

Proof.

Example 1. Make a change of variables that transforms the quadratic form $Q(\mathbf{x}) = x_1^2 - 8x_1x_2 - 5x_2^2$ into a quadratic form with no cross-product term.

Definition. A quadratic form $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ is classified as one of the following:

- (1) **positive definite** if _____ for all $\mathbf{x} \neq \mathbf{0}$.
- (2) **positive semidefinite** if _____ for all $\mathbf{x}$.
- (3) **negative definite** if _____ for all $\mathbf{x} \neq \mathbf{0}$
- (4) **negative semidefinite** if _____ for all $\mathbf{x}$
- (5) **indefinite** if $Q(\mathbf{x})$ takes on both _____ and _____ values

A symmetric matrix A has the same classification as the associated quadratic form $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$.

THEOREM Let A be an $n \times n$ symmetric matrix. The quadratic form $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ is

- (a) positive definite if and only if all of the _____ of A are _____
- (b) positive semidefinite if and only if all of the eigenvalues of A are _____
- (c) negative definite if and only if all of the eigenvalues of A are _____
- (d) negative semidefinite if and only if all of the eigenvalues of A are _____
- (e) indefinite if and only if A has both _____ and _____ eigenvalues

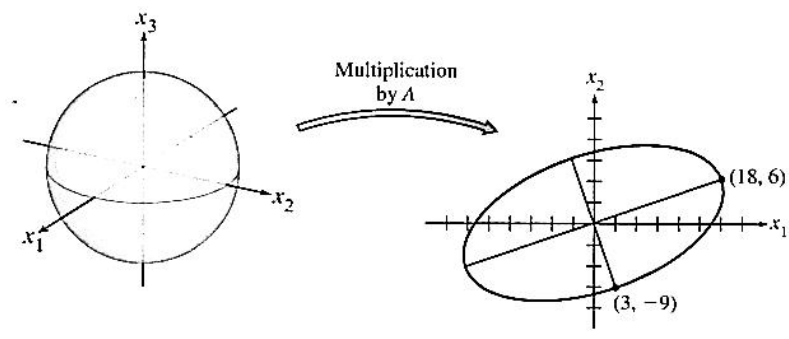
THEOREM Let $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ be a quadratic form with associated $n \times n$ symmetric matrix A . Let the eigenvalues of A be $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$. Then the following are true, subject to the constraint $\|\mathbf{x}\| = 1$

- (a) $\lambda_1 \geq Q(\mathbf{x}) \geq \lambda_n$
- (b) The maximum value of $Q(\mathbf{x})$ is λ_1 and it occurs when $\mathbf{x}$ is a unit eigenvector corresponding to λ_1 .
- (c) The minimum value of $Q(\mathbf{x})$ is λ_n and it occurs when $\mathbf{x}$ is a unit eigenvector corresponding to λ_n .

Proof:

Geometry of Eigenvalues for Square Matrix

Example 2. Let $A = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}$, Then, the linear transformation $T(\mathbf{x}) = A\mathbf{x}$ maps the unit sphere $\{\mathbf{x} : \|\mathbf{x}\| = 1\}$ onto an ellipse as shown below.



Find a unit vector $\mathbf{x}$ at which the length $\|A\mathbf{x}\|$ is maximized and compute this maximum length.

Singular Values of an $m \times n$ Matrix

Definition. If A is an $m \times n$ matrix, the **singular values** of A are the _____ roots of the _____ of $A^T A$ and are denoted by $\sigma_1, \dots, \sigma_n$. It is conventional to arrange the singular values so that $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n$

Example 3. Find the singular values of $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$.

Example 4. Find the singular values of $A = \begin{bmatrix} 4 & 6 \\ 0 & 4 \end{bmatrix}$.